DRAFT: Analysis of Geomorphic Metrics for OSAM Revisions

This document summarizes the analysis of geomorphic metrics for OSAM revisions. Since the work is on-going, all information is preliminary. This document addresses three items: (1) the geomorphic metrics being collected, including definitions and collection methods, (2) the stratification variables under evaluation, and (3) the procedures for developing reference and regional curves.

Geomorphic Metrics

The following list includes the collected metrics that can align with geomorphology in typical stream quantification tools. These metrics will all be statistically analyzed and considered for reference curve development. No decisions will be made on which metrics/reference curves to include into OSAM. The final selection of metrics and reference curves to be included in the revised OSAM will be by OEPA and USACE with support from stakeholders.

Table 1. Geomorphic metrics included in data collection and to be analyzed across sites.

Metric Name	Definition / Formula	Data Collection Method
Bankfull Width/Depth Ratio	Bankfull Width / Mean Bankfull Depth	Cross-section survey at representative riffle
Bank Height Ratio	Low Bank Height / Max Bankfull Depth	Cross-section survey; low bank height and maximum bankfull depth measured at each cross-section.
Entrenchment Ratio	Flood-Prone Area Width / Bankfull Width	Cross-section survey; flood-prone area width measured at twice maximum bankfull depth elevation.
Pool to Pool Spacing Ratio	Pool Spacing / Bankfull Width	Field measured along longitudinal profile; distance between pools divided by bankfull width.
Pool Depth Ratio	Pool Depth / Mean Bankfull Depth	Cross-section measurements at pool features; maximum pool depth compared to mean bankfull depth.
Percent Riffle	Riffle Length / Reach Length	Geomorphic feature mapping along the reach; riffle lengths summed and divided by total reach length.
Aggradation Ratio	Width-depth ratio (W/D) compared reference W/D	Width-depth ratio collected by field survey. Reference width-depth ratio from revised regional curves.
Sinuosity	Reach Length/Valley Length	Field survey and GIS calculation
Large Woody Debris (LWD) frequency	Count of LWD pieces within the 100m segment with the most LWD	Field count along the reach; the 100m segment with the highest LWD density is used.
Riparian Buffer Width (ft)	Width of natural vegetated buffer from top of bank to lateral drainage divide	Desktop delineation using aerial imagery and LiDAR, verified on-site. Measured perpendicular to the valley fall-line on both sides of the channel.
Concentrated Flow Points	Count of ephemeral erosional features that concentrate runoff directly into the stream	Field observation and mapping along the reach.
Percent Streambank Erosion	Eroding streambank length / Total streambank length	Field assessment of both banks along the reach; eroding lengths identified and measured.

Woody Stem Basal Area (m^2/ha)	Sum of basal areas of all native woody species ≥3.0 in (7.6 cm) DBH per hectare	Measured within 30-ft radius vegetation plots; DBH recorded at 4.5 ft (1.37 m) for all qualifying native woody stems.
Canopy Cover (%)	Aerial cover of native woody vegetation >16.4 ft (5 m) tall within the riparian buffer	Visual estimate within 30-ft radius vegetation plots using standardized cover class ranges and midpoints.
BEHI / NBS	Bank Erosion Hazard Index and Near Bank Stress ratings (Rosgen method)	Multiple BEHI/NBS assessments along the reach; dominant condition reported.
Native Herbaceous Cover (%) L&R	Aerial cover of native herbaceous vegetation <3.28 ft (1 m) tall	Visual estimate within 1 m^2 nested subplots at each vegetation plot using standardized cover class ranges and midpoints.
Invasive Cover (%) L&R	Aerial coverage of all invasive and non-native vegetation (herbaceous, shrub, and tree)	Visual estimate within each 30-ft radius vegetation plot using standardized cover class ranges.
Native Shrub Cover (%) L&R	Aerial cover of native woody species 3.28 to 16.4 ft (1 to 5 m) tall	Visual estimate within 30-ft radius vegetation plots using standardized cover class ranges and midpoints.
Grain Size (Particle Size Analysis)	Pebble count distribution (e.g., D16, D50, D84, D95)	Wolman pebble count at representative riffle(s); minimum 100 particles measured.

Stratification

Each metric from Table 1 will be evaluated for whether grouping data by selected variables results in statistically distinct distributions that improve curve performance. Statistical methods under consideration include Kruskal-Wallis tests, Wilcoxon rank-sum tests, ordinary least squares regression, and negative binomial regression depending on the model fit and type of data being evaluated. Model fit must be balanced with stability, as excessive stratification reduces sample size and can produce unstable curves.

Table 2. Stratification variables being tested across metrics.

Stratification Variable	Description	Applied To
EPA Level III Ecoregion	Eastern Corn Belt Plains (ECBP), Interior Plateau (IP), Huron/Erie Lake Plains (HELP), Western Allegheny Plateau (WAP), Erie Drift Plains (EDP), Eastern Great Lakes Lowlands (EGLL).	All metrics
Drainage Area Category	Category 1: ≤1 mi^2; Category 2: >1 to 20 mi^2; Category 3: >20 mi^2.	All metrics
Rosgen Stream Classification	Stream type (A, B, C, E, etc.) based on channel morphology (Rosgen 1996). Used in the MNSQT, USGS Ohio Regional Curves, and other SQTs.	Metric-by-metric
Valley Type	Confined, semi-confined, or unconfined alluvial valleys. Used for riparian metrics in other SQTs.	Metric-by-metric
Bed Material / Grain Size	D50 grain size classes; used as an explanatory variable in Sherwood & Huitger (2005).	Metric-by-metric
Other Variables	Examples could include stream slope, sinuosity, stream order, thermal regime, and watershed land use characteristics. Additional variables may be explored.	Metric-by-metric

Procedure for Reference Curves Development

Reference curves translate field-measured metric values into dimensionless index scores from 0.0 to 1.0: Functioning (0.70 to 1.00), Functioning-at-Risk (0.30 to 0.70), and Not Functioning (0.00 to 0.30). These procedures follow standard accepted practices for SQTs and are informed by the Minnesota SQT Science Document (Kieser & Associates et al. 2020). This process is iterative because ecoregion data are still being collected. The general steps are as follows:

1) **Select regionally relevant parameters and metrics.** Metric selection was completed jointly by Ohio EPA and the Cheetah Team.

2) **Review available data sources and collect reference data.** Approximately 90 reference sites across six ecoregions are being sampled in accordance with QAPP protocols (Ohio EPA 2025). Data sources are evaluated using the five USEPA (2003) assessment factors and include: (a) published literature (e.g., Sherwood and Huitger 2005), (b) existing regional and national datasets (e.g., NRSA, USGS regional curves), (c) Ohio-specific field data, and (d) expert opinion. Quality assurance procedures include verifying bankfull identification, confirming Bank Height Ratio ≤1.1 per QAPP criteria, and screening for outliers.

3) **Define reference context and condition gradients.** Statistical analysis describes variation among reference sites but does not define degraded conditions. For scoring, three functional capacity categories are applied: Functioning, At-Risk, and Non-functioning, consistent with other Stream Quantification Tools. See Table 3 for detailed explanation of these functional categories. Reference sites are evaluated using K-means clustering and principal component analysis to examine variability, identify outliers, and assess the need for stratification.

4) **Identify thresholds and fit reference curves.** Thresholds are informed by: (a) published literature, (b) existing or newly developed regional or national datasets, and (c) expert opinion, consistent with other SQTs. Stratification variables are tested by investigating which independent variables provide significant explanatory power through ordinary least squares regression for continuous data, poisson or negative binomial regression for count data such as large wood (LWD). Kruskal-Wallis and Wilcoxon rank-sum are used as well to test significant differences when investigating categorical variables of interest such as Ecoregions, drainage area category, or Rosgen stream type. For both stratified and non-stratified metrics, the interquartile range of reference standard sites defines the 0.70 to 1.00 scoring range. Linear relationships will be used for the final reference curves.

   Building the lower end of the curve: Lower-end curve development requires additional information because reference data alone cannot define non-functioning conditions. Approaches under consideration include: (a) interpolation below the reference distribution, (b) comparison with comparable NRSA metrics where methods align, (c) published threshold values, and (d) collection of data from degraded streams. Advisory team input is particularly valuable for this step.

5) **Evaluate metric independence and redundancy.** Principal component and correlation analyses identify unique versus redundant metrics to maximize information while minimizing overlap (Kieser & Associates et al. 2020). After analysis, OEPA and the project team will determine the final set of metrics.

6) **Incorporate reference curves into scoring framework.** Each metric is converted to a 0 to 1 index score using a reference curve. Metric scores are combined into parameter scores, parameter scores into functional category scores, and functional category scores into an overall reach condition score (Kieser

& Associates et al. 2020). Final scoring weights and structure for OSAM will be determined by OEPA and the project team.

7) **Documentation and peer review.** Supporting documentation will be prepared, consistent with other Stream Quantification Tools, and all analyses will undergo review.

Table 3. Functional capacity definitions used to define threshold values and develop reference curves for the MN SQT (Kieser & Associates 2020).

Functional Capacity	Definition	Index Value Range
Functioning	A functioning value means that the metric is quantifying or describing the functional capacity of one aspect of a function-based parameter in a way that **supports aquatic ecosystem structure and function.** The reference standard concept aligns with the definition for a reference condition for biological integrity (Stoddard et al. 2006). A score of 1.00 represents an un-altered or pristine condition (native or natural condition). A range of index values (0.70-1.00) is used for characterizing reference standard to account for natural variability, recognizing that reference standard datasets include sites that reflect least disturbed condition (i.e., the best available conditions given current anthropogenic influence per Stoddard et al. 2006).	0.70 to 1.00
Functioning-at-risk	A functioning-at-risk value means that the metric is quantifying or describing one aspect of a function-based parameter in a way that **may support aquatic ecosystem structure and function**, but not at a reference standard level**.** In many cases, this indicates the parameter is adjusting in response to changes in the reach or the catchment towards lower or higher function. This range characterizes a grey area, where a resource is neither achieving a reference standard nor is significantly degraded or impaired.	0.30 to 0.69
Not-functioning	A not functioning value means that the metric is quantifying or describing one aspect of a function-based parameter in a way that **does not support aquatic ecosystem structure and function.** An index value less than 0.30 represents an impaired or severely altered condition relative to reference standard, and an index value of 0.00 represents a condition that provides no functional capacity for that metric. Index values of 0.00 are often the minimum value possible for a metric.	0.00 to 0.29

Procedure for Regional (Hydraulic Geometry) Curves Development

Regional curves relate bankfull channel dimensions, including width, mean depth, and cross-sectional area, to drainage area using power-function relationships (to characterize non-linear relationships). In this analysis, regional curves are being primarily used to help screen reference sites. Sites that fall well outside the regional curve may not represent reference conditions and may be removed from the reference dataset. Independent of OSAM, regional curves can show bankfull dimensions compared to what is expected for its drainage area, which can help identify departure from reference conditions. The 2005 USGS Ohio Regional Curves (Sherwood and Huitger 2005) defined two bankfull regions, including a separate Interior Plateau region. This analysis evaluates whether updated datasets and ecoregion-based stratification improve predictive performance. The general steps are as follows:

1) **Collect bankfull geometry data.** Bankfull width, mean depth, cross-sectional area, and flood-prone area width are measured in accordance with QAPP protocols (Ohio EPA 2025) and methods consistent with Harrelson et al. (1994).

2) **Determine drainage area and verify bankfull identification.** Drainage area is delineated using GIS and LIDAR-based watershed analysis. Bankfull indicators were identified in the field and will be cross-checked using Bank Height Ratio criteria. Sites with uncertain bankfull identification are removed from analysis.

3) **Develop regression equations.** Log-log regression of bankfull dimensions versus drainage area is used to derive power-function relationships of the form Dimension = a × DA^b. Coefficients of determination, standard errors, and confidence intervals are calculated (Sherwood and Huitger 2005; Leopold and Maddock 1953).

4) **Test stratification and explanatory variables.** Stratification by ecoregion, the original two-region classification, drainage area category, and Rosgen stream type will be tested. Multiple regression models incorporating slope, median bed material size (D50), and mean elevation will be considered if they meaningfully improve model performance. Integration of published site data from Sherwood (2005) may be considered to improve site representation across ecoregions.

5) **Benchmark against published curves and documentation.** Newly developed curves are compared with Sherwood and Huitger (2005) and other published relationships. Final equations, stratification decisions, standard errors, and comparison plots will be documented and peer-reviewed.

References Cited

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